14/8/2024 (AN)

Hall Ticket Number:

MR-21

Code: B314CB6S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July/August-2024

Computer Organization and Architecture

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Draw the instruction word format and indicate the number of bits in each part.	2	1	1	1
1. b)	Give examples for conditional and unconditional transfer instructions.	2	1	1	1
1. c)	What do you mean by microprogram? What are the different types of microprograms?	2	2	1	1
1. d)	Differentiate between a direct and an indirect address instruction?	2	2	2	1
1. e)	Perform 2's complement of (i) 10010010 (ii) 01101100.	2	1	5	1
1. f)	Define normalization in floating point representation.	2	2	5	1
1. g)	Why are read and write control lines in a DMA controller bidirectional?	2	1	3	3
1. h)	An address space is specified by 24bits and the corresponding memory space by 16 bits. How many words are there in the address space?	2	2	3	2
1. i)	How do branch instructions cause problems in a pipelined processor?	2	1	4	2
1. j)	List out replacement algorithms.	2	1	4	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain about various functional units of a basic computer with the help of block diagram.	5	2	2	1
2. b)	Design the 4-bit register transfer to common bus using multiplexers.	5	1	2	1
	OR				
2. c)	Explain about various conditional branch instructions in a general-purpose computer.	5	2	2	2
2. d)	What is an interrupt? Explain briefly about different program interrupts and what are the sequence of events involved in handling an IRQ?	5	1	2	2
3. a)	Explain various instruction formats with examples.	5	2	1	1
3. b)	Explain with the help of diagram the register organization of computer with memory bank.	5	2	1	1
	OR				
3. c)	What do you mean by addressing mode? Analyse the following addressing modes. i) Index addressing mode ii) Relative addressing mode iii) Immediate addressing mode iv) Implied addressing mode.	5	4	1	1
3. d)	Compare hardwired control with micro programmed control. Is it possible to have a hardwired control associated with control memory?	5	3	1	1
4. a)	What is floating point representation? Explain the IEEE standard for floating point representation with examples.	5	1	5	3
4. b)	Explain about Booth's Multiplier with flowchart.	5	3	5	3

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	OR				
4. c)	Build an algorithm in flowchart form for adding and subtracting two fixed point binary numbers when negative numbers are in signed 1's complement representation.	5	5	5	3
4. d)	Construct a Decimal Arithmetic Unit to perform BCD addition and BCD Subtraction.	5	3	5	3
5. a)	With the help of block diagram of interface between processor and peripheral devices explain its operations.	5	3	3	2
5. b)	Describe the DMA operation with neat diagram.	5	3	3	1
	OR				
5. c)	List out different types of memory. Explain the functionality of RAM and ROM chips with the help of function table.	5	1	3	1
5. d)	Distinguish between Direct mapping, Set-Associative mapping and Associate mapping of cache memory.	5	4	3	1
6. a)	Illustrate between Array processor and Vector Processor. State the advantages, disadvantages and applications of vector processors.	5	2	4	1
6. b)	Compare and contrast between parallel processing and pipeline processing with suitable examples.	5	2	4	1
	OR				
6. c)	What is pipelining? Demonstrate the pipeline organization with an example.	5	2	4	2
6. d)	Analyze the concurrent access to memory and cache coherency.	5	4	4	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome